ICAD RT-5 · Set 3 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
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18	a	b	c	d
19	a	b	c	d
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Numerical — write the value

21	
22	
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Chemistry

1	a	b	c	d
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3	a	b	c	d
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20	a	b	c	d

Numerical — write the value

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Mathematics

1	a	b	c	d
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3	a	b	c	d
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19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
22	
23	
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Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 3

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

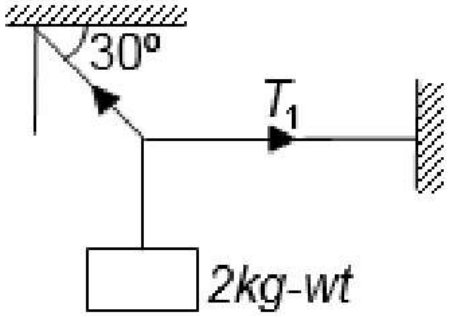
PART A · Q1–20 · SINGLE CORRECT

1. P4L1V5Q10 A particle moves on a straight line whose velocity and displacement relation is $v^2 = 400 - 4x$. Which of the following statements is/are correct? (one or more options may be correct) (a) The time in which particle stops is 10 s (b) The distance travelled by particle in last 2 s before coming to rest is 4 m (c) This is an example of non-uniformly accelerated motion (d) This is an example of uniformly accelerated motion

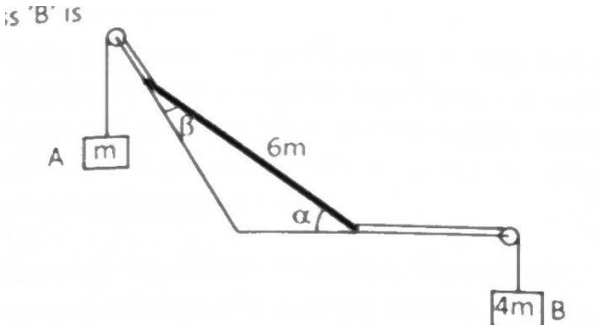
2. P4L5V4Q2 Displacement $s = at + bt^2 - ct^3$ (a, b, c constants). Velocity when acceleration is zero: (a) $a+b^2/c$ (b) $a+b^2/2c$ (c) $a+b^2/3c$ (d) $a+b^2/4c$

3. P4L12VSuggQ5 An airplane pilot wants to fly from city A to city B which is 1000 km due north of city A. The speed of the plane in still air is 500 km/hr. The pilot neglects the effect of the wind and directs his plane due north and 2 hours later finds himself 300 km due north-east of city B. The wind velocity is (a) 150 km/hr at 45° N of E (b) 106 km/hr at 45° N of E (c) 150 km/hr at 45° N of W (d) 106 km/hr at 45° N of W

4. P5XVRT3Q19 A body of weight 2 kg is suspended as shown in the figure. The tension T_1 in the horizontal string (in kg wt) is (a) $2/\sqrt{3}$ (b) $\sqrt{3}/2$ (c) $2\sqrt{3}$ (d) 2

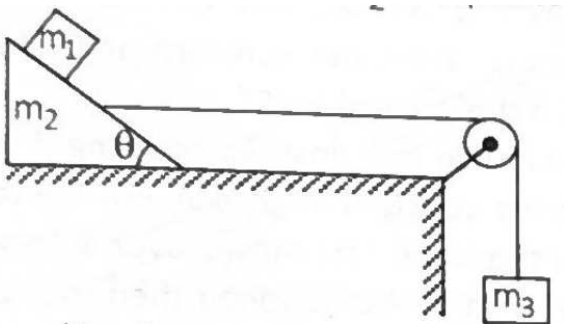


5. P6L4V6Q2 Two blocks 'A' and 'B' of mass 'm' and '4m' are connected with a rod PQ of length 'l' and mass 6m through light strings. There is no friction anywhere. End P of the rod slides on the incline (its string runs up along the incline over a pulley to A) and end Q slides on the horizontal floor (its string runs horizontally over a pulley to B). If the system is released from rest and the rod PQ slides on the incline, then for the position shown in the figure (β = angle between rod and incline at P, α = angle between rod and floor at Q), the ratio of the speeds of the mass 'A' to that of the mass 'B' is (a) $\sin\alpha/\sin\beta$ (b) $\cos\alpha/\cos\beta$ (c) $\cos(\alpha+\beta)/\cos(\alpha-\beta)$ (d) $\sin(\alpha+\beta)/\sin(\alpha-\beta)$



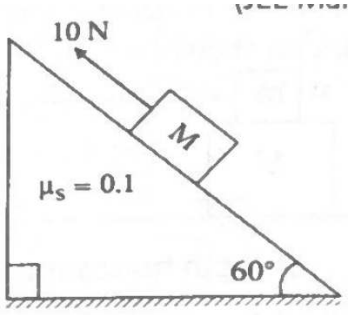
6. P6L7V3Q4 The ratio of the weight of a man in a stationary lift and in a lift accelerating downwards with a uniform acceleration 'a' is 3 : 2. The acceleration of the lift is (a) g/3 (b) g/2 (c) g (d) 2g

7. P6L7V5Q6 A block m_1 rests on the inclined face (angle θ) of a wedge m_2 that lies on a smooth horizontal table; the wedge is pulled by a light string passing over a smooth pulley at the table edge to a hanging mass m_3 (all surfaces smooth). The value of m_3 such that it is just able to prevent relative motion of m_1 with respect to m_2 is equal to (a) $(m_1 + m_2)\tan\theta/(1 - \tan\theta)$ (b) $(m_1 + m_2)\cot\theta/(1 - \cot\theta)$ (c) $(m_1 + m_2)\sin\theta/(1 - \sin\theta)$ (d) $(m_1 + m_2)\cos\theta/(1 - \cos\theta)$



8. P7L1V5Q2 A 3 kg block on a rough horizontal table ($\mu_s = 0.60$, $\mu_k = 0.45$; wood on brick) is pulled by a force that ramps linearly from 0 to 30 N in 5 s. When does it start moving? ($g = 9.8\ m/s^2$) (a) At 2.9 s; N = mg and F = μsN decide onset. (b) Never; μs increases with time. (c) At 1.5 s; sliding starts earlier than limiting friction. (d) At 5.0 s by definition of ramp end.

9. P7SBVPYQQ7 A block of mass 1 kg is pushed up a surface inclined at 60° to the horizontal by a force of 10 N parallel to the inclined surface, with μ s = 0.1. When the block is pushed up by 10 m along the inclined surface, the work done against frictional force is ($g = 10\ m\ s^{-2}$): (a) $5\sqrt{3}$ J (b) 5 J (c) 5×10^3 J (d) 10 J

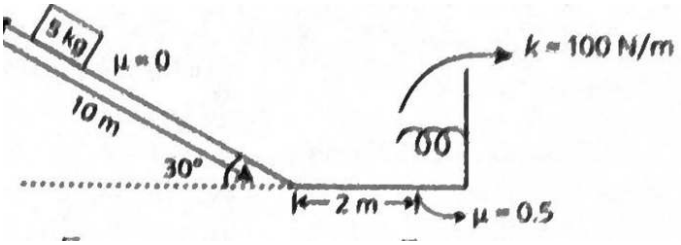


10. P8L7V5Q7 A small block of mass 100 g is pressed against a horizontal spring fixed at one end to compress the spring through 5.0 cm (in figure). The spring constant is 100 N/m. When released, the block moves horizontally till it leaves the spring. The spring is on a smooth table top at A, 2 m above the ground at B. The horizontal distance BC from the free end of the spring where the block will hit the ground is (g = 10 m/s²) (a) 0.5 m (b) 1 m (c) 0.75 m (d) None of these

onstant k, the other end of sprin
The block can move on a horizo
e coefficient of friction between
ace is μ. then the compression of
maximum extension of the spr
aximum compression is
(b) $\frac{mg\mu}{k}$

11. P8L8V5Q10 (one or more options may be correct) In the following K = kinetic energy, U = potential energy, and W = work. In using the expression $K_2 + U_2 = K_1 + U_1 + W_other$, we recognise that (a) we are assuming no friction is present (b) gravitational and elastic potential energies would be included in U_1 and U_2 (c) K and U can be either positive or negative, depending on the reference point chosen (d) W_other can be positive or negative

12. P8SBVPYQQ9 A 5 kg block is simply released from the top of an inclined plane as shown in the figure: it slides 10 m down a frictionless (μ = 0) 30° incline, then travels 2 m along a rough horizontal stretch with μ = 0.5, and then hits a spring of constant k = 100 N/m (the surface under the spring is frictionless). The maximum compression in the spring when the block hits the spring is (g = 10 m/s²) (a) √6 m (b) √5 m (c) 1 m (d) 2 m



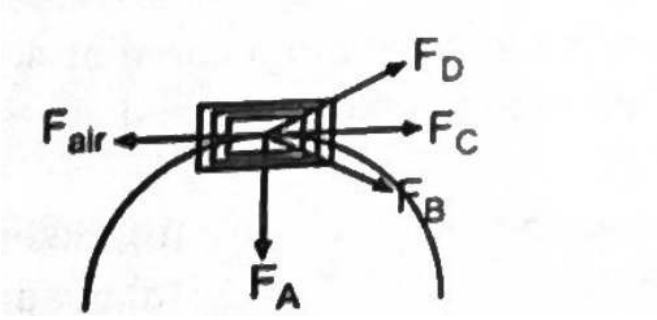
13. P9L5L6V6Q10 A particle of mass m moving in the x direction with speed 2v is hit by another particle of mass 2m moving in the y direction with speed v. If the collision is perfectly inelastic, the percentage loss in the energy during the collision is close to (a) 44% (b) 50% (c) 56% (d) 62%

14. P9SBVPYQQ10 A body of mass M at rest explodes into three pieces, in the ratio of masses 1:1:2. Two smaller pieces fly off perpendicular to each other with velocities of 30 m s^{-1} and 40 m s^{-1} respectively. The velocity of the third piece will be (a) 35 m s^{-1} (b) 50 m s^{-1} (c) 25 m s^{-1} (d) 15 m s^{-1}

15. P10L2V6Q7 The tap in a standard audio cassette has a length L = 120 m; the tape plays running at constant linear speed for t = 2 hr. As the tape starts, the full reel has an outer radius of R 1 = 12√2 cm, and an inner radius of R 2 = 5√2 cm. At some point during the play, both reels have the same angular speed. Calculate this angular speed in rad/s. (a) 1/500 (b) 5/39 (c) 1/(510√2) (d) None of these

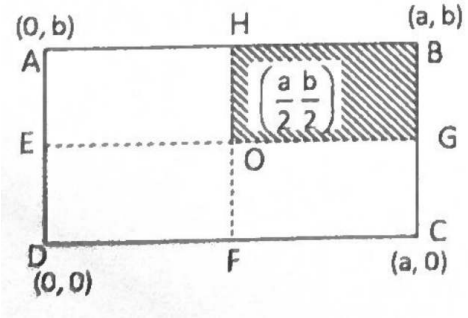
16. P10L4V4Q5 A simple pendulum consists of a light string from which a spherical bob of mass M is suspended. The distance between the point of suspension and centre of the bob is l. The bob is given a tangential velocity v at the equilibrium position (i.e., at the lowest point). If v = √(2gl), the angle described by the string going from one extreme position to the other will be: (a) π/4 (b) π/2 (c) 3π/2 (d) π

17. P10L6L7V3Q3 A car travels with constant speed on a circular road on level ground. In the diagram, F air is the force of air resistance on the car. Which of the other forces shown best represents the horizontal force of the road on the car's tires? (a) F A (b) F B (c) F C (d) F D

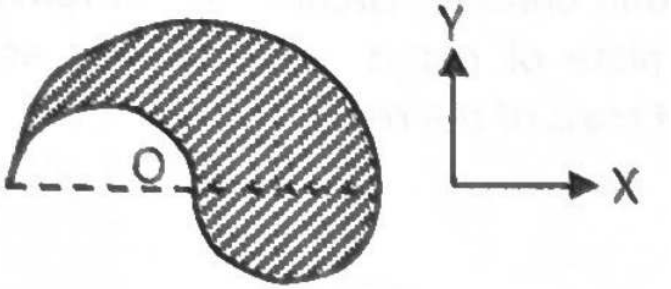


18. P11L1V2Q1 Three masses are placed on the x–axis : 300 g at origin, 500 g at x = 40 cm and 400 g at x = 70 cm. The distance of the centre of mass from the origin is : (a) 40 cm (b) 45 cm (c) 50 cm (d) 30 cm

19. P11L3V6Q5 A uniform rectangular thin sheet ABCD of mass M has length a and breadth b, as shown in the figure. If the shaded portion HBGO is cut-off, the coordinates of the centre of mass of the remaining portion will be (a) (2a/3, 2b/3) (b) (5a/12, 5b/12) (c) (3a/4, 3b/4) (d) (5a/3, 5b/3) [Figure: D(0,0), C(a,0), B(a,b), A(0,b); O is the centre (a/2, b/2), H the mid-point of AB and G the mid-point of BC, so HBGO is the top-right quarter.]



20. P11L3V6Q9 From a uniform semi-circular thin disc of radius R, a semi-circular portion of radius R/2 is removed and placed in a new position as shown in figure. Taking O as origin and X and Y-axis as shown in figure, centre of mass of new setup is at (a) (R/4, 4R/3π) (b) (−R/4, R/3π) (c) (R/4, R/π) (d) (R/4, R/3π) [Figure: the original half-disc occupies y > 0 with its diameter along the x-axis from −R to +R; the half-disc of radius R/2 standing on the left half of that diameter is removed and re-attached below the diameter on the right half, giving the S-shaped figure.]



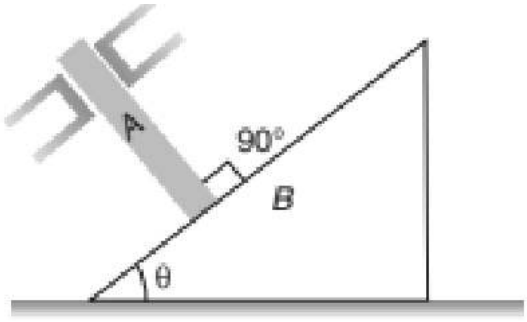
PART B · Q21–25 · NUMERICAL VALUE

21. P3XVRT1Q25 The resultant of two vectors A and B is perpendicular to A and its magnitude is half that of B. The angle between A and B is ____ °.

22. P4XVCAT3Q25 An object moving with a speed of 6.25 m/s, is decelerated at a rate given by $dv/dt = -2.5\sqrt{v}$ where v is the instantaneous speed. The time (in s) taken by the object, to come to rest, would be

23. P6XVRT4Q11 A monkey of mass 30 kg climbs a rope which can withstand a maximum tension of 360 N. The maximum acceleration which this rope can tolerate for the climbing of monkey is : (g = 10 m/s²) (A) 2 m/s² (B) 3 m/s² (C) 4 m/s² (D) 5 m/s²

24. P6XVRT4Q22 In the figure shown the relation between magnitudes of a_A and a_B is related as a_A = n a_B sinθ, then n is



25. P10XVCAT6Q20 A coin placed on a rotating turntable just slips if it is placed at a distance of 4 cm from the centre. If the angular velocity of the turntable is doubled, it will just slip at a distance of (A) 1 cm (B) 2 cm (C) 4 cm (D) 8 cm

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L6V5Q3 Total energy of one mole of an ideal monoatomic gas at 27 °C is (take R ≈ 2 cal mol^{-1} K^{-1}): (a) 600 cal (b) 900 cal (c) 800 cal (d) 300 cal

2. C2L2VSuggQ4 The emission spectrum of hydrogen always shows discrete bright lines rather than a continuous spread of colors. Which statement best explains this? (a) Atoms emit all possible frequencies continuously. (b) Electrons in atoms can occupy only discrete energy levels, giving fixed energy transitions. (c) Continuous radiation is absorbed by the atom. (d) Spectra arise due to interference between waves.

3. C2L3VSuggQ2 A metal has threshold frequency $\nu_0 = 6.0 \times 10^{14}$ Hz. Light of frequency $\nu = 8.0 \times 10^{14}$ Hz shines on it. Using Einstein's equation K_max = h(ν − ν0), what is (ν − ν0)? (a) 2.0×10^{14} Hz (b) 1.5×10^{14} Hz (c) 8.0×10^{14} Hz (d) 6.0×10^{14} Hz

4. C3L4VSuggQ4 A compound has an empirical formula mass of 29. Its molecular mass is 58. What is its molecular formula? (a) Same as empirical formula (b) 2 × empirical formula (c) 4 × empirical formula (d) Cannot be determined

5. C4L1V5Q4 The elements which are characterised by the outer shell configuration ns^1 to ns^2np^5 are collectively called (a) Transition elements (b) Representative elements (c) Lanthanides (d) Inner transition elements

6. C4L3V5Q5 The ionic radii of Li^+, Be^{2+} and B^{3+} follow the order: (a) $Be^{2+} > B^{3+} > Li^+$ (b) $Li^+ > B^{3+} > Be^{2+}$ (c) $B^{3+} > Be^{2+} > Li^+$ (d) $Li^+ > Be^{2+} > B^{3+}$

7. C4SBVPYQQ53 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. Assertion A: The energy required to form Mg 2+ from Mg is much higher than that required to produce Mg^+. Reason R: Mg 2+ is small ion and carry more charge than Mg^+. In the light of the above statements, choose the correct answer from the options given below. (a) Both A and R are true but R is NOT the correct explanation of A (b) A is true but R is false (c) A is false but R is true (d) Both A and R are true and R is the correct explanation of A

8. C5L4V5Q10 The molecule of CO_2 has angle 180°. It can be explained on the basis of: (a) sp^3-hybridisation (b) sp-hybridisation (c) sp^2-hybridisation (d) d^2sp^3-hybridisation

9. C5L7VSuggQ3 Which of the following overlaps can produce a σ molecular orbital? (a) p–p sidewise overlap (b) s–p sideways overlap (c) p–p head-on overlap (d) p–d sideways overlap

10. C5L8VSuggQ4 Which of the following best defines an antibonding molecular orbital, according to Molecular Orbital Theory? (a) An orbital formed by constructive overlap of atomic orbitals, increasing stability. (b) An orbital formed by destructive overlap of atomic orbitals, with a node between nuclei. (c) An orbital that always contains paired electrons. (d) An orbital that lowers the potential energy of the molecule.

11. C5SBVPYQQ45 Number of molecules/ions from the following in which the central atom is involved in sp^3 hybridization is NO_3^-, BCl_3, ClO_2^-, ClO_3^- (a) 4 (b) 3 (c) 2 (d) 1

12. C5XVCAT5Q40 If an electron is added to N_2 to form the ion N_2^-, in which molecular orbital will the electron be placed? (a) σ2s (b) σ2s (c) $\pi 2p_x$ (d) σ*2p_z

13. C6L2V3Q5 Heat flows from one system to another primarily due to a difference in: (a) Pressure (b) Volume (c) Temperature (d) Internal energy

14. C6L2V6Q6 [One or more than one option may be correct] Which of the following statements about internal energy are correct? (one or more options may be correct) (a) Internal energy is an extensive property (b) Internal energy is a state property (c) Absolute value of internal energy can be directly measured (d) Internal energy depends on quantity of substance and state variables

15. C6L5V5Q5 Which are correct to express work terms? (a) Work = Capacity factor × Intensity factor; where capacity factor is a measure of extent of work done and intensity factor is a measure of force responsible for work (b) Electrical work = E×nF; E is intensity factor; nF is capacity factor (c) Expansion work = P×ΔV; P is intensity factor; ΔV is capacity factor (d) All of these

16. C6L9V6Q2 The standard enthalpy of formation ($\Delta_f H^\circ$) at 298 K for methane $CH_4(g)$ is −74.8 kJ mol −1 . The additional information required to determine the average energy for C—H bond formation would be: (a) the dissociation energy of H_2 and enthalpy of sublimation of carbon (b) latent heat of vaporization of methane (c) the first four ionization energies of carbon and electron gain enthalpy of hydrogen (d) the dissociation energy of hydrogen molecule, H_2

17. C6L11V5Q2 The incorrect expression among the following is: (a) In isothermal process, W reversible = $-nRT \ln(V_f/V_i)$ (b) $\ln K = (\Delta H^\circ - T\Delta S^\circ)/RT$ (c) $K = e^{-\Delta G^\circ/RT}$ (d) $\Delta G_{system}/\Delta S_{total} = -T$

18. C6L11V5Q7 Passage (6–8). A particular reaction has a negative value for the free energy change. Then at ordinary temperature (a) It has a large (−ve) value for the entropy change (b) It has large (+ve) value for enthalpy change (c) It has a small (+ve) value for enthalpy change (d) It has a (+ve) value for the entropy change and a (−ve) value for enthalpy change

19. C6L12V5Q10 Which of the following are true about resonance energy? (Heats of formation are taken as heat evolved, i.e. positive for an exothermic formation.) (one or more options may be correct) (a) Resonance energy = Experimental heat of formation − calculated heat of formation. (b) Resonance energy = Calculated heat of formation − Experimental heat of formation. (c) Greater the resonance energy, more the compound will be stable. (d) Lesser the resonance energy, more the compound will be stable

20. C6XVCAT6Q31 The enthalpy of hydrogenation of $C_6H_6(l)$ is −205 kJ and resonance energy of $C_6H_6(l)$ is −152 kJ/mole, then enthalpy of hydrogenation of [cyclohexene, structure shown] is [Given ΔH_{vap} of $C_6H_{10}(l)$, $C_6H_6(l)$, $C_6H_{12}(l)$ all are equal] (a) −519 kJ/mole (b) −229 kJ/mole (c) −619 kJ/mole (d) −119 kJ/mole



PART B · Q21–25 · NUMERICAL VALUE

21. C1SBVPYQQ10 At constant temperature, a gas is at a pressure of 940.3 mm Hg. The pressure at which its volume decreases by 40% is _____ mm Hg

22. C2XVCAT3Q49 What is the maximum total number of electrons that can be accommodated in all the possible sub-shells having n + l = 4?

23. C3XVRT2Q46 Aqueous urea solution is 20% by mass of solution. Calculate the percentage by mass of solvent.

24. C3XVRT3Q33 What is the mass of carbon dioxide (CO_2) produced when 5 moles of methane (CH_4) react completely with oxygen gas (O_2) in the balanced chemical equation: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$? (Molar mass: CH_4 = 16 g/mol, O_2 = 32 g/mol, CO_2 = 44 g/mol) (A) 44 grams (B) 88 grams (C) 176 grams (D) 220 grams

25. C4XVRT4Q26 The atomic numbers of four elements are given below. Which one is an alkaline earth metal? (A) 10 (B) 20 (C) 30 (D) 40

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M3SBVPYQQ3 The least value of $(\cos^2\theta - 6\sin\theta\cos\theta + 3\sin^2\theta + 2)$ is (a) −1 (b) $4 + \sqrt{10}$ (c) 1 (d) $4 - \sqrt{10}$

2. M4L1V4Q4 If $(\alpha - 0)/1 = (\beta - 0)/1 = -2(-1)/(1^2 + 1^2)$, then the value of $\alpha + \beta$ is (a) 1 (b) 2 (c) −1 (d) 0

3. M4L2VSuggQ4 The function f(x) = [x] (greatest integer function) is defined for all real x. Which of the following describes its range? (a) All positive integers (b) All real numbers (c) All integers (d) All non-negative integers

4. M4XVRT4Q62 f(x) = $x^3 - 6x^2 + 6x$, then f(2^(1/3) + 2^(2/3) + 2) has the value (a) 2 (b) 4 (c) 6 (d) 8

5. M5AL1V5Q5 If sin10° = p and cos70° = (√(1−p^2) − xp)/2, then the value of $(2x^2 + 1)$ is: (a) 3 (b) 7 (c) 8 (d) 10

6. M6L1V2Q3 Solve: If 5 < x, then what is x + 3? (a) x + 3 < 8 (b) x + 3 > 8 (c) x + 3 = 8 (d) x + 3 < 5

7. M6L3V5Q6 The sum of all real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is: (a) 7 (b) 4 (c) 1 (d) 5

8. M7L4V6Q2 The set of real values of x for which 2^(log_√2(x−1)) > x + 5, is: (a) (−∞, −1) ∪ (4, ∞) (b) (4, ∞) (c) (−1, 4) (d) none of these

9. M7XVRT3Q53 The equation $5^{(1 + \log_5 \cos x)} = 5/2$ has (a) no solution (b) one solution (c) two solutions (d) more than two solutions

10. M8AL3V6Q6 A person is to count 4500 currency notes. Let a n denote the number of notes he counts in the n th minute. If $a_1 = a_2 = \ldots = a_{10} = 150$ and $a_{10}, a_{11}, \ldots$ are in an A.P. with common difference −2, then the time taken by him to count all notes is (a) 24 minutes (b) 34 minutes (c) 125 minutes (d) 135 minutes

11. M8AXVCAT5Q64 Let $\Sigma_{k=1}^{n} a_k = \alpha n^2 + \beta n$. If $a_{10} = 59$ and $a_6 = 7a_1$, then α + β is equal to (a) 12 (b) 3 (c) 5 (d) 7

12. M8BL1V6Q2 The sum of 3 decreasing numbers in AP is 27. If −1, −1, 3 are added to them respectively, the resulting series is in G.P. The numbers are (a) 5, 9, 13 (b) 15, 9, 3 (c) 13, 9, 5 (d) 17, 9, 1

13. M8DL2V5Q2 Find sum of the following series upto infinite terms 1 + 4/3 + $7/3^2$ + $10/3^3$ + $13/3^4$ + ...∞ terms (a) 15/7 (b) 15/2 (c) 15/4 (d) None of these

14. M8DL3V6Q6 Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. If a < b < c and a + b + c = 3/2, then the value of a is (a) 1/(2√2) (b) 1/(2√3) (c) 1/2 − 1/√3 (d) 1/2 − 1/√2

15. M8DSBVPYQQ5 Consider the sequence $a_1, a_2, a_3, \ldots$ such that $a_1 = 1$, $a_2 = 2$ and a_(n+2) = 2/a_(n+1) + a_n for n = 1, 2, 3, ... If $[(a_1 + 1/a_2)/a_3]\cdot[(a_2 + 1/a_3)/a_4]\cdot[(a_3 + 1/a_4)/a_5]\cdot\ldots\cdot[(a_{30} + 1/a_{31})/a_{32}] = 2^{(\alpha)}\cdot({}^{61}C_{31})$, then α is equal to (a) −30 (b) −31 (c) −60 (d) −61

16. M9L2V4Q4 If (x + iy)(2 - 3i) = 4 + i then (x, y) = (a) (1, 1/13) (b) (-5/13, 14/13) (c) (5/13, 14/13) (d) (-5/13, -14/13)

17. M9L3V6Q4 If a + ib + ci^2 + di^3 = (x + iy)^2 then √(a - ib + ci^2 - di^3) is equal to (a) (-x + iy)i (b) -x - iy (c) x - iy (d) x + iy

18. M9L4V2Q4 Find $|z_1| + |z_2|$ for $z_1 = 1 + i$ and $z_2 = 2 + i$. (a) √2 + √5 (b) √3 (c) 3 (d) 2

19. M9L5V2Q2 The principal argument of z = -3 + 3i (a) π/4 (b) -π/4 (c) 3π/4 (d) -3π/4

20. M9L6V1Q2 Argument θ of z = x + iy is: (a) $\tan^{-1}(x/y)$ (b) $\tan^{-1}(y/x)$ (c) $\cos^{-1}(x)$ (d) $\sin^{-1}(y)$

PART B · Q21–25 · NUMERICAL VALUE

21. M2XVRT4Q75 A college awarded 38 medals in football, 15 in basketball and 20 in cricket. If the medals went to a total of 58 men and only three men got medals in all three sports, then the number of men received medals in exactly two of the three sports is

22. M5CXVRT2Q66 If cosθ = 1/2(a + 1/a) and cos3θ = λ·1/2(a^3 + 1/a^3), then λ = (A) 1/4 (B) 1/2 (C) 1 (D) none

23. M7L2V4Q3 The value of x which satisfies 2·3^(x+1) − 6·3^(x−1) − 3^x = 9 is

24. M7XVCAT4Q72 If the solution set of (1.25)^(1 − x) < (0.64)^(2(1 + √x)) is (5^a, ∞) then value of a is

25. M8BSBVPYQQ11 Let $a_1, a_2, a_3, \ldots$ be a GP of increasing positive numbers. If the product of fourth and sixth terms is 9 and the sum of fifth and seventh terms is 24, then $a_1a_9 + a_2a_4a_9 + a_5 + a_7$ is equal to _____.